Individual Contribution Report

Student Name: Mark Ellery Contreras
Group: SureballNa (MiniOSNova)
Assigned Role: Project Leader

What did I contribute to the project?

- Coordinated the group members, assigned module responsibilities, and tracked project progress to meet our submission requirements.
- Set up and managed the official GitHub repository (`MiniOSNova`) and connected it to Visual Studio Code.
- Combined all individual Python modules into the main program and verified that the final version ran smoothly without errors.
- Assisted in drafting the User Manual and Technical Documentation, and tested the system commands in the terminal.

What features did I personally develop?

- Designed the custom startup banner and the main `while True` loop for the command prompt.
- Implemented the input parser to separate commands from arguments and route them to the appropriate modules.
- Programmed the core general commands: `help`, `clear`, `time`, `systeminfo`, and `exit`.
- Added CLI error handling to prevent crashes from missing arguments or unknown commands.

What problems did I encounter?

- GitHub Authentication Error (403 Permission Denied) because a classmate's account was cached in Windows Credential Manager.
- Git Merge Conflict when our local and remote branches got out of sync.
- The program crashed with an `IndexError` when commands like `run` or `mkdir` were typed without a name.

How did I solve those problems?

- I removed the old GitHub credentials from Windows Credential Manager and logged in with my own account.
- I aborted the stuck Git merge and force-pushed our clean local code to GitHub.
- I added an `if len(user_input) > 1:` check to make sure the program warns the user instead of crashing when arguments are missing.

What did I learn from this project?

I learned how an operating system's command shell continuously listens for user input, parses commands, and delegates tasks to specialized modules. I also gained hands-on experience managing a team repository with Git and GitHub in VS Code, resolving merge conflicts, and organizing technical documentation.

Evidence of my contribution:

- **GitHub Repository Commits:** Commit history under `markelleryc-cpu` in https://github.com/markelleryc-cpu/MiniOSNova.
- **Source Code:** Complete implementation of `main.py` in the project repository.
- **Project Documentation:** Exported PDFs in the `docs/` folder and terminal test screenshots.

